

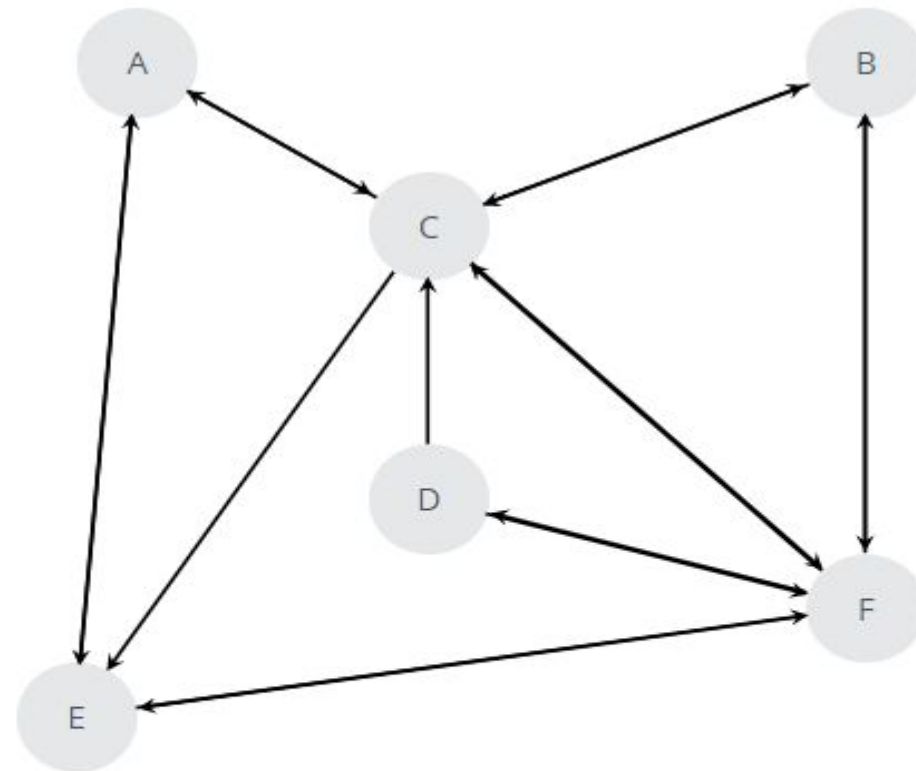


ค่ายโอลิมปิกวิชาการ ครั้งที่ 18 ศูนย์มหาวิทยาลัยเทคโนโลยีพระจอมเกล้า พระนครเหนือ

DFS

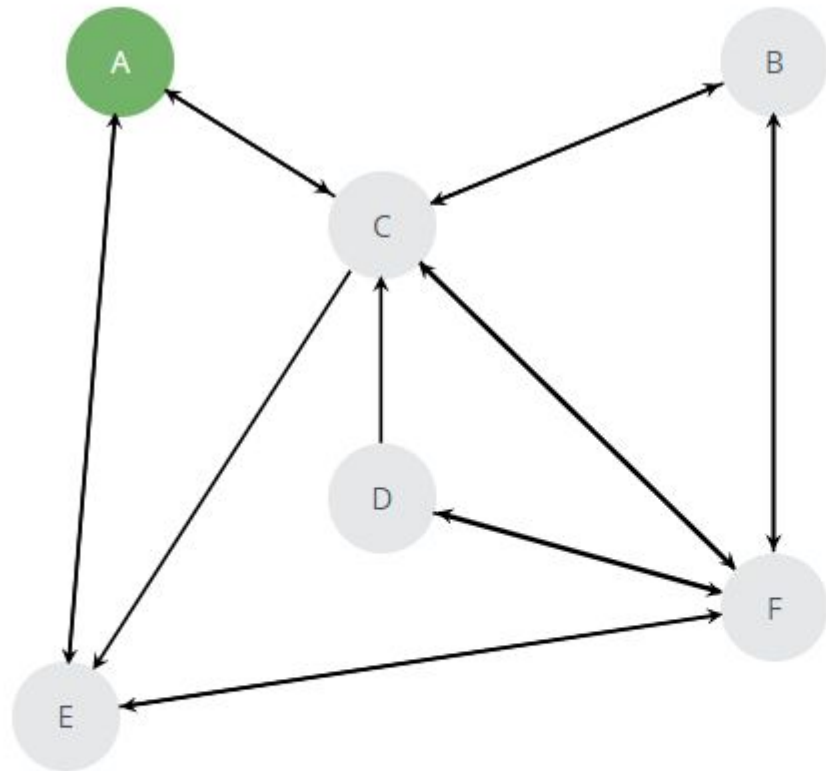
ภาควิชาวิทยาการคอมพิวเตอร์และสารสนเทศ มหาวิทยาลัย

มหาวิทยาลัยเทคโนโลยีพระจอมเกล้าพระนครเหนือ



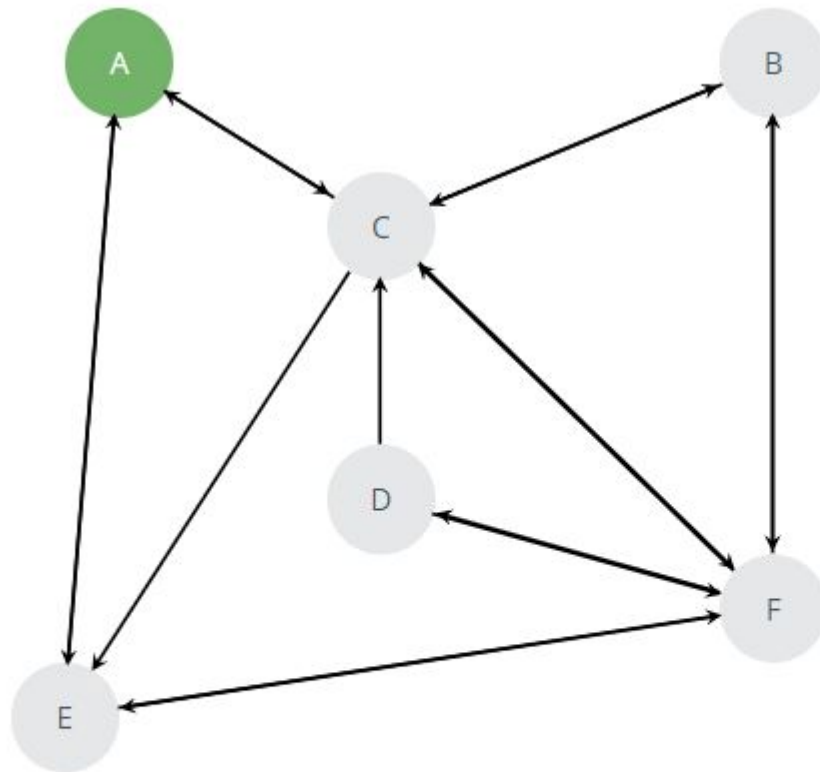
Steps:

Let us look at the details of how a depth-first search works.

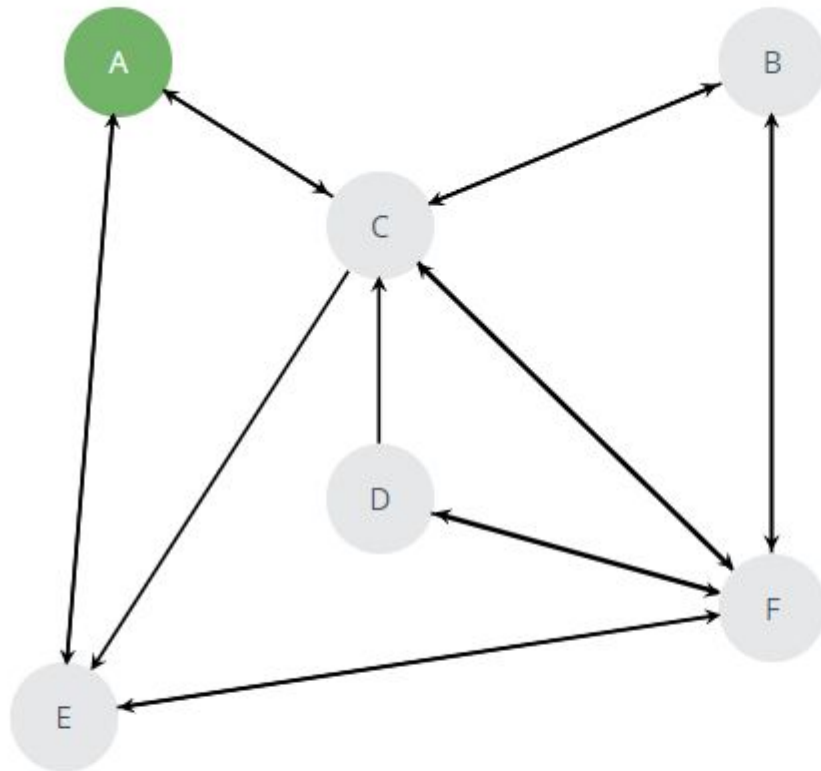


Steps:

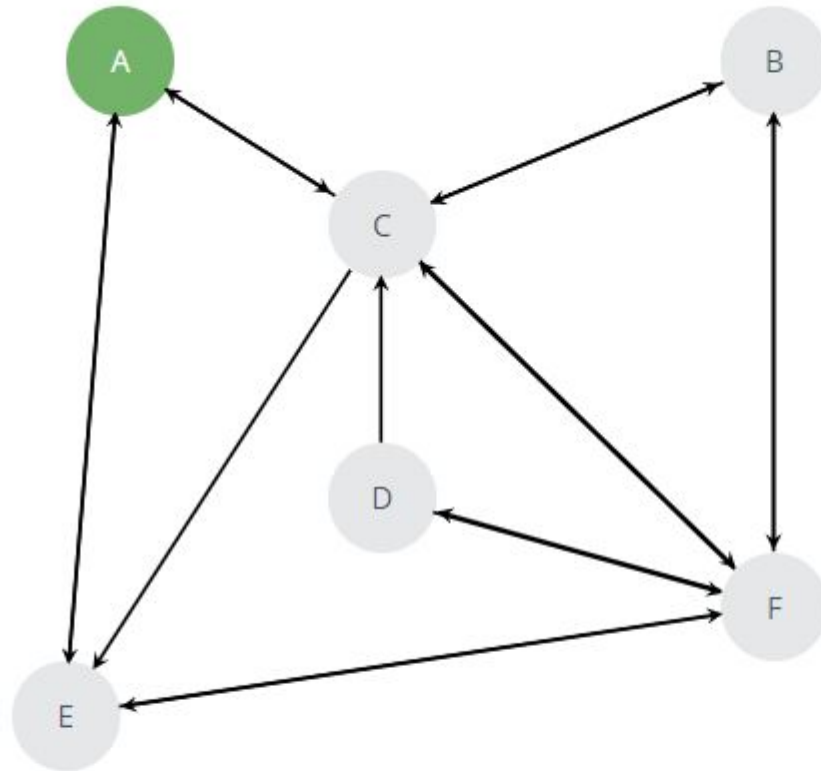
Mark node A



Steps:
Add A to the stack

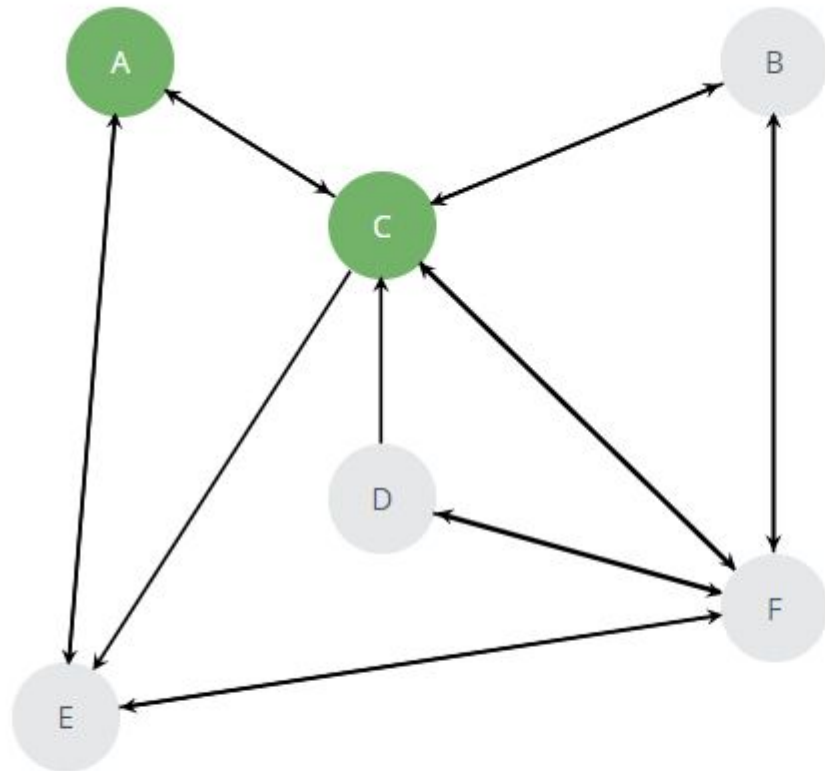


Steps:
Process (A,C)

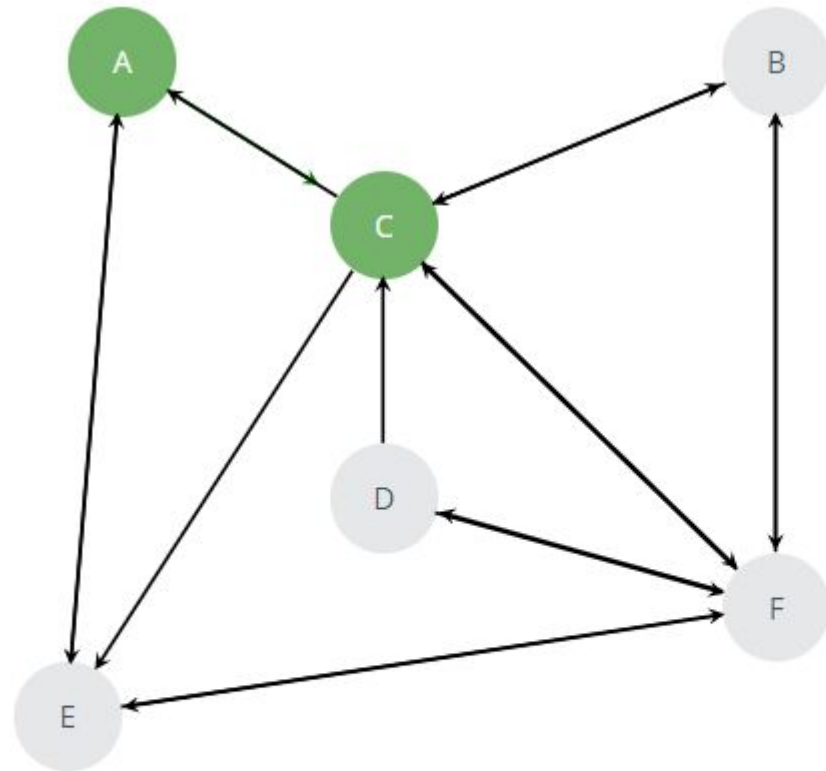


Steps:

Print (A,C) and call depth first search on C

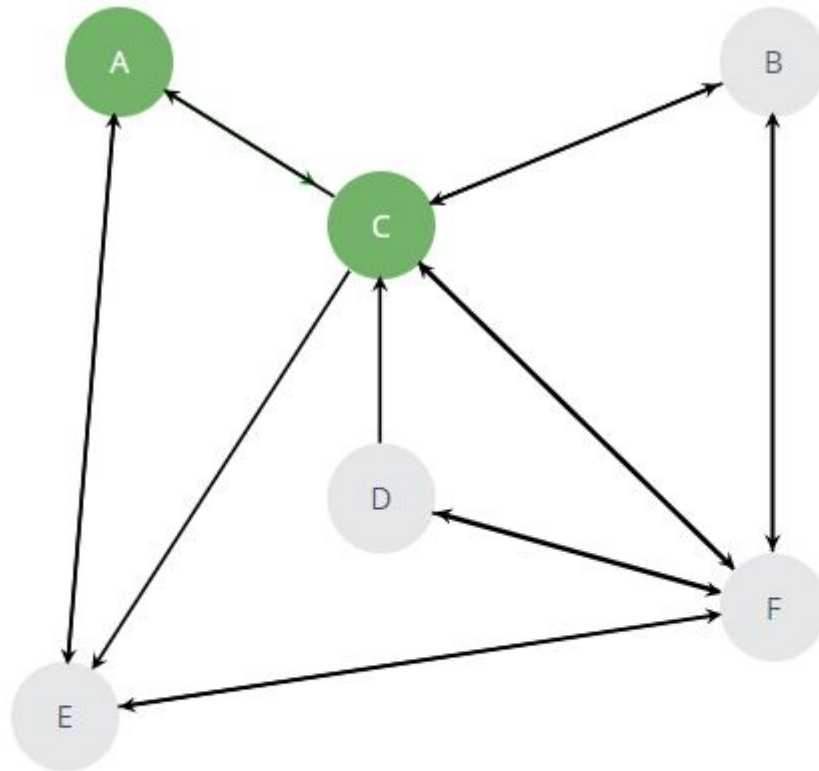


Steps:
Mark node C



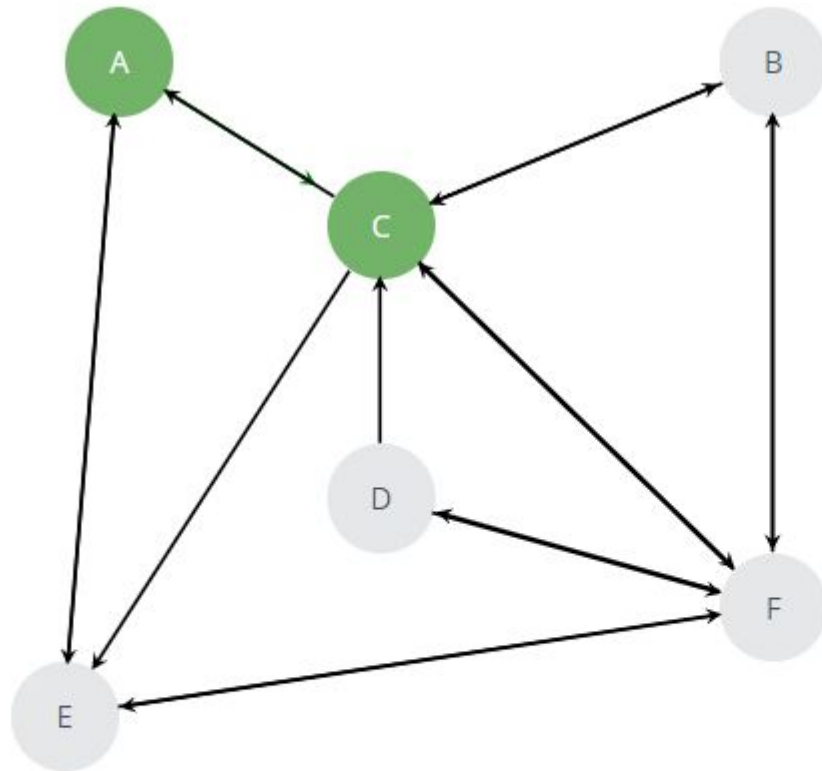
Steps:

Add C to the stack

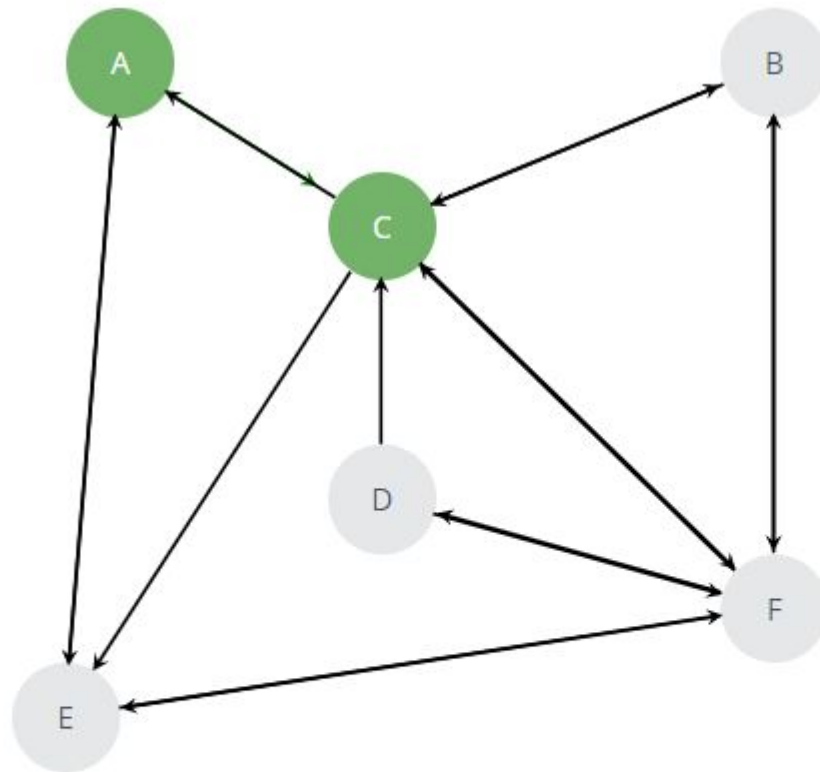


Steps:

Node A already marked

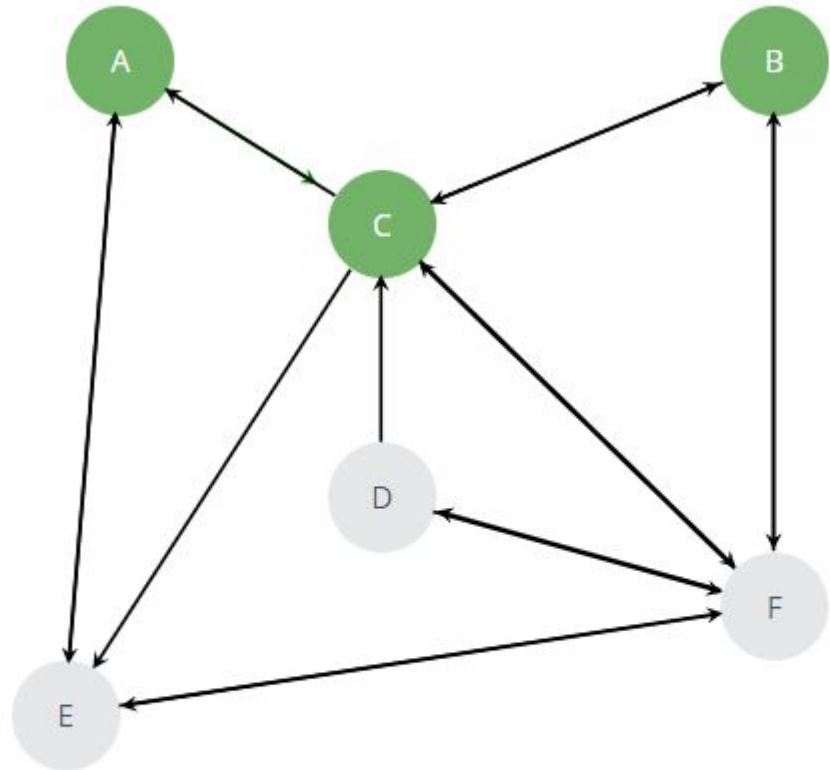


Steps:
Process (C,B)

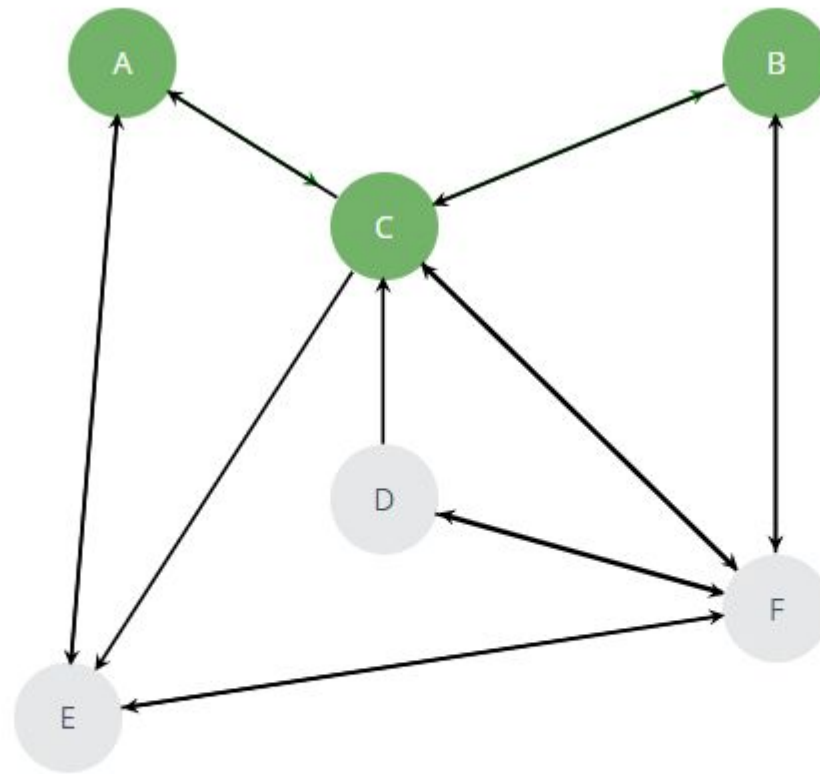


Steps:

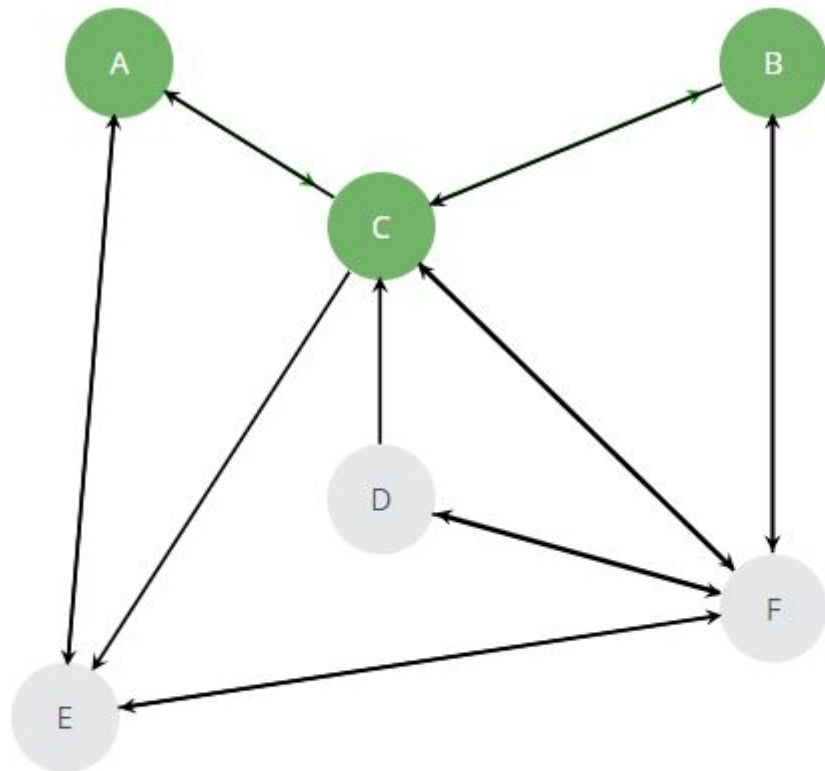
Print (C,B) and call depth first search on B



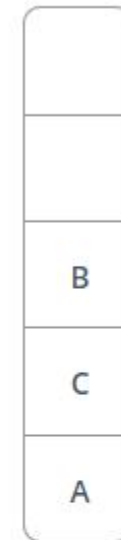
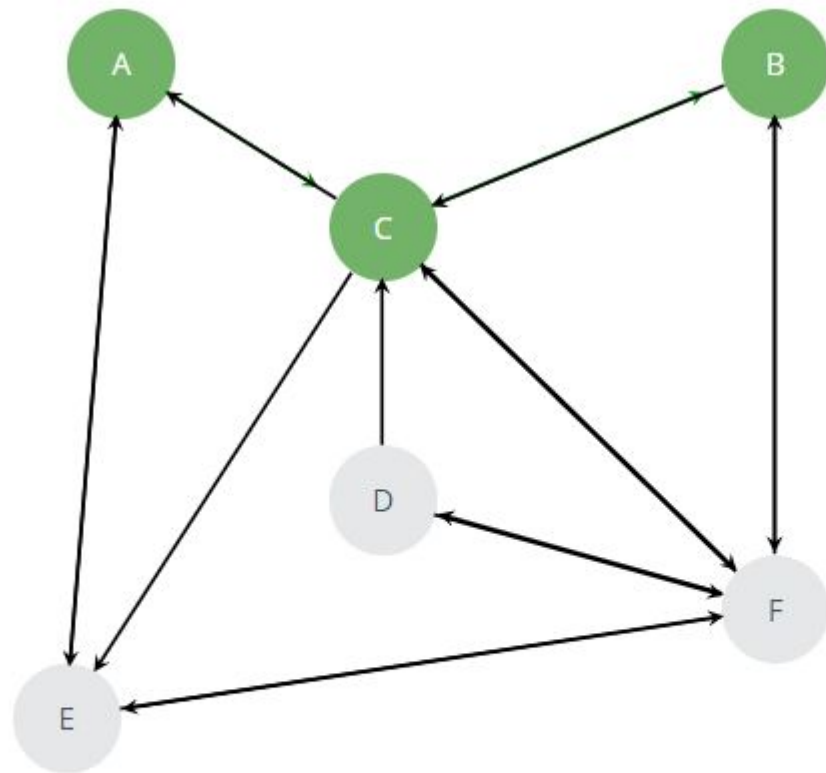
Steps:
Mark node B



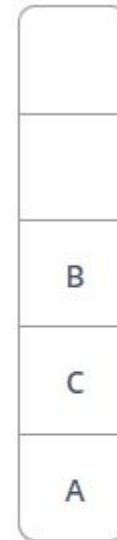
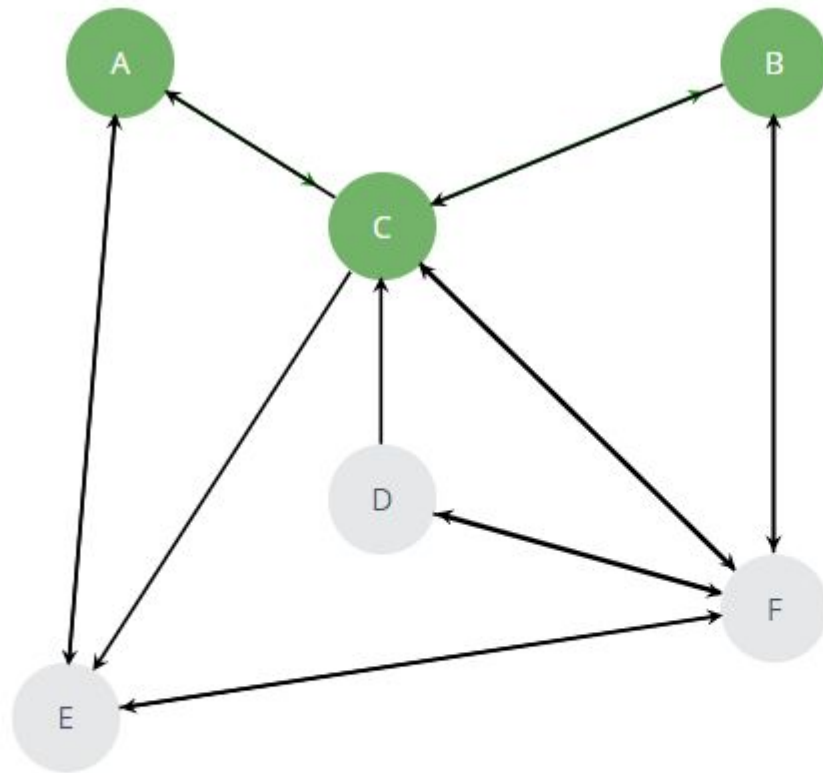
Steps:
Add B to the stack



Steps:
Node C already marked

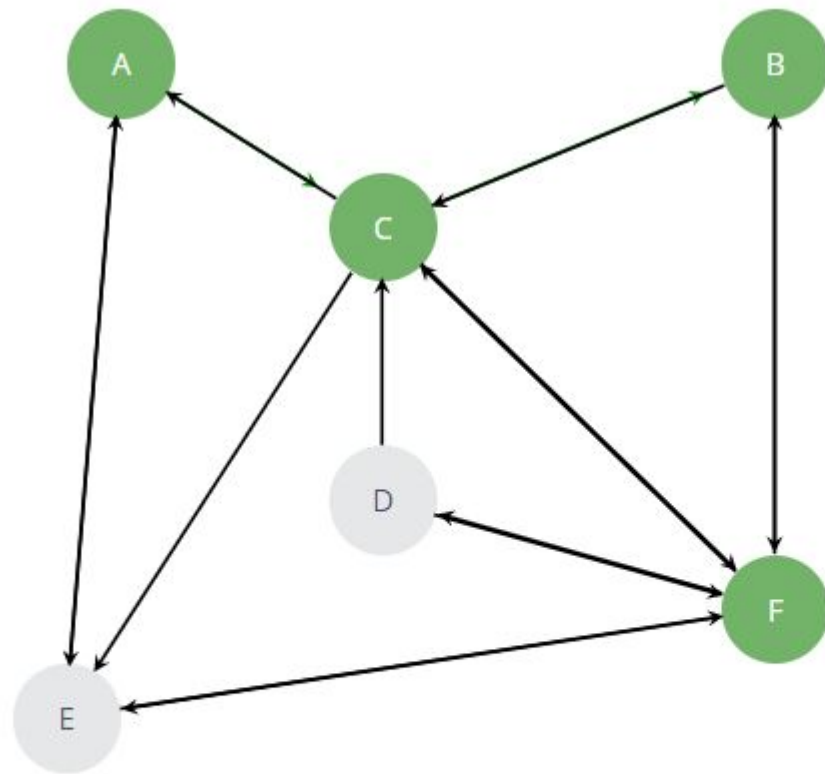


Steps:
Process (B,F)

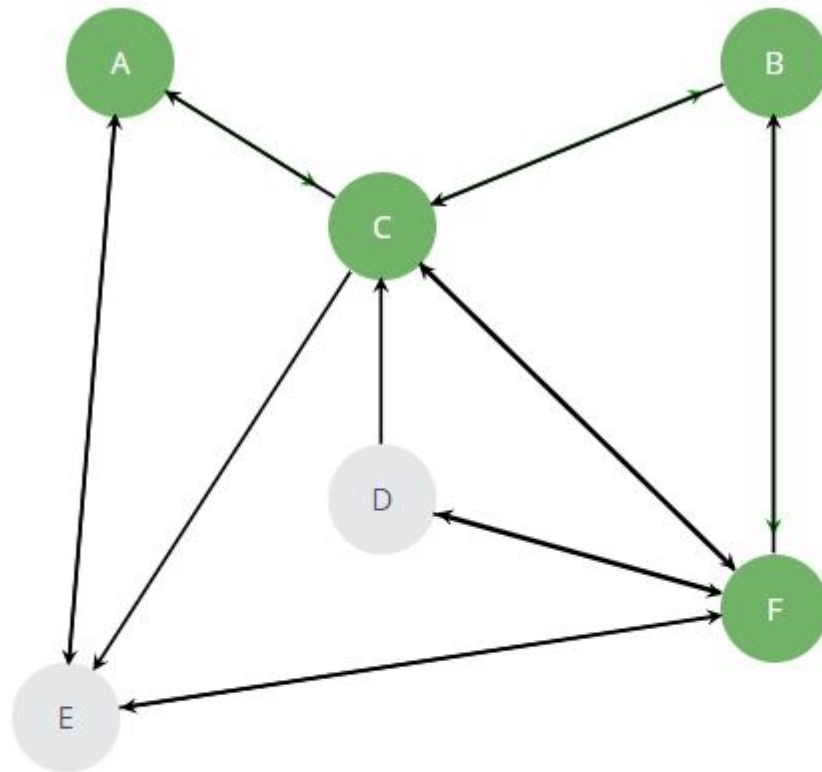


Steps:

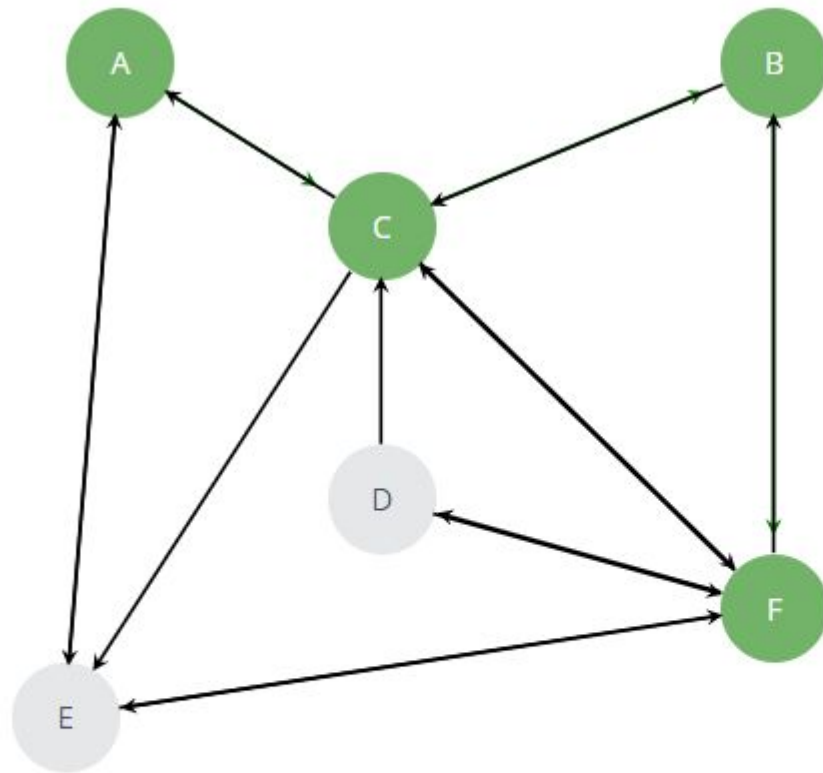
Print (B,F) and call depth first search on F



Steps:
Mark node F

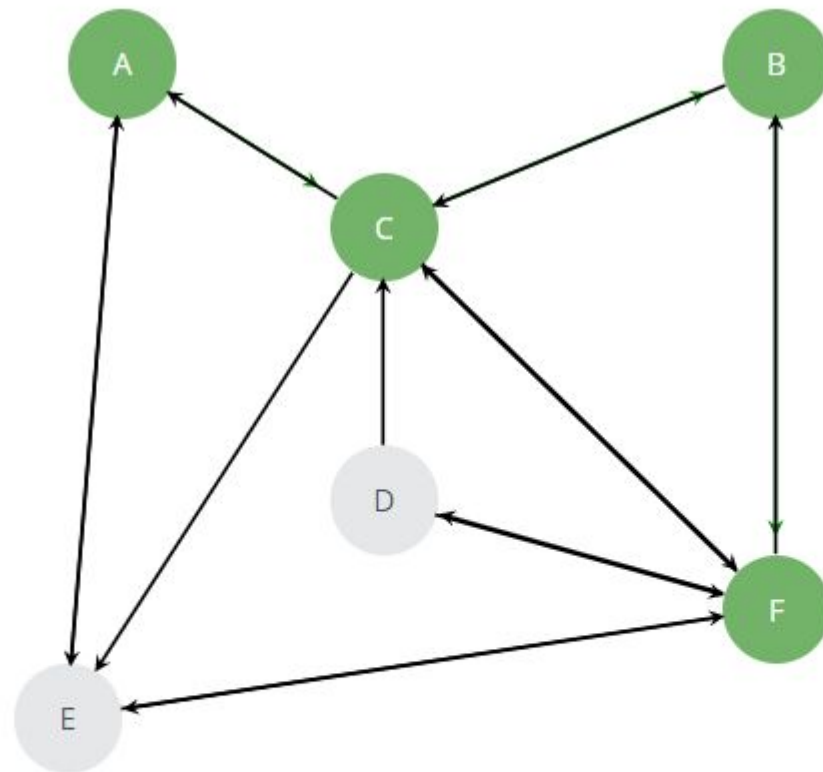


Steps:
Add F to the stack

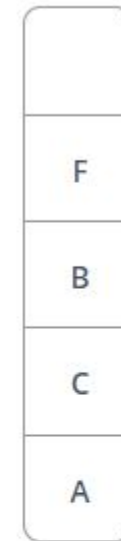
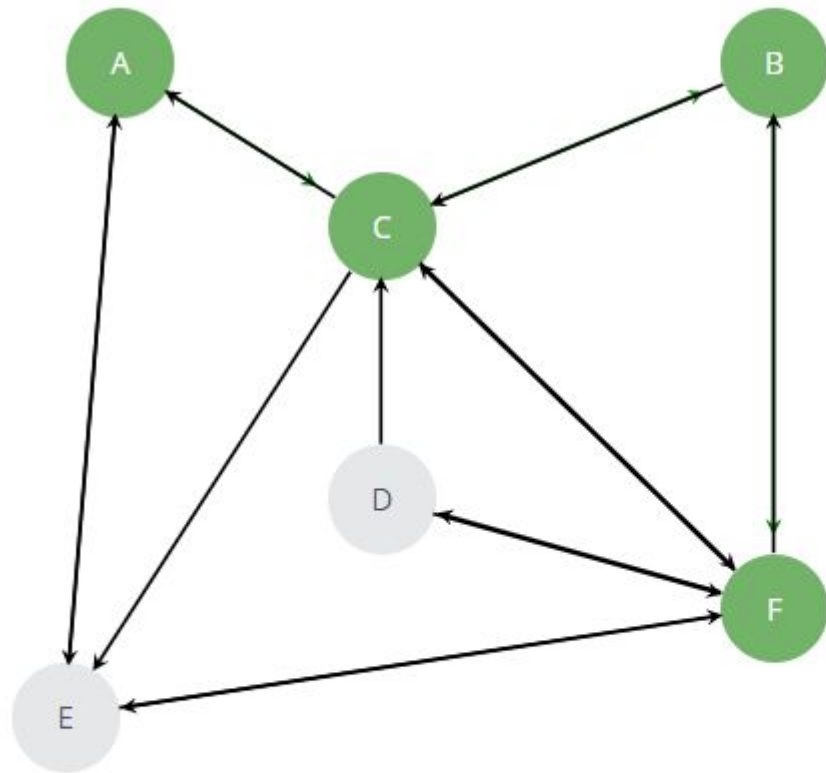


Steps:

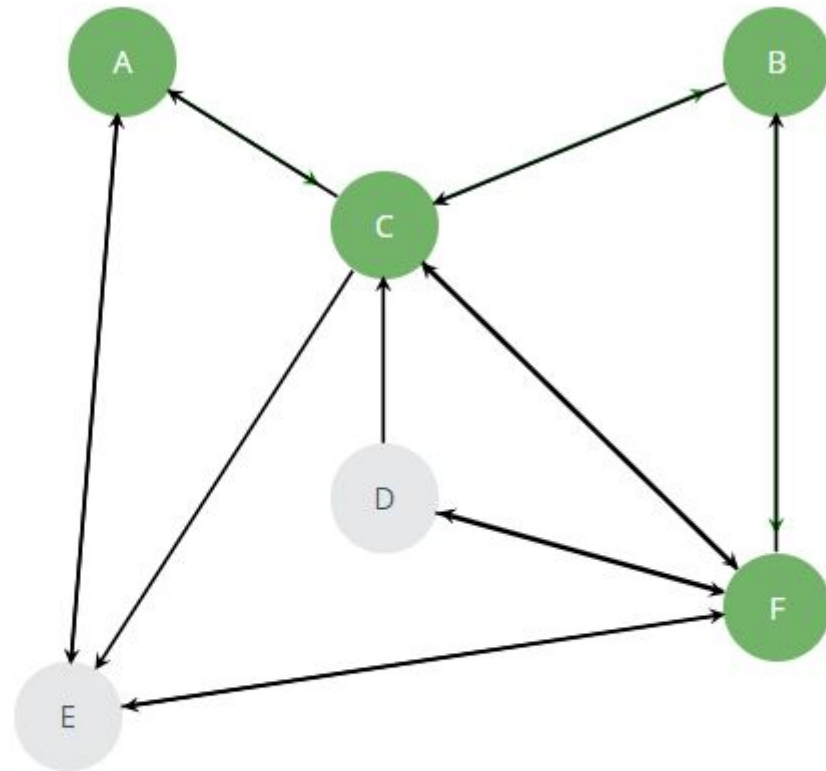
Node B already marked



Steps:
Node C already marked



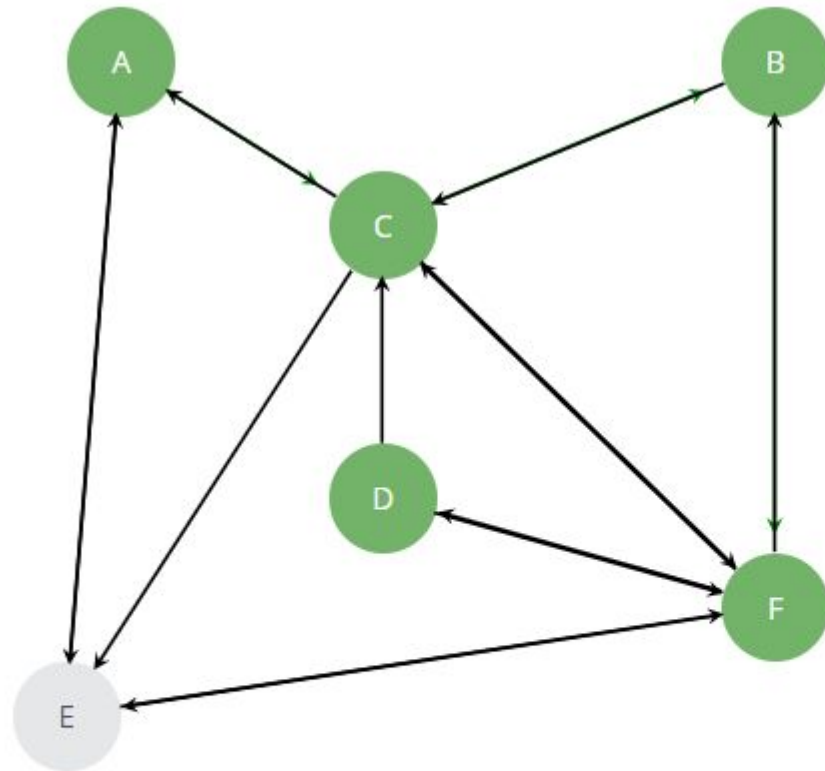
Steps:
Process (F,D)



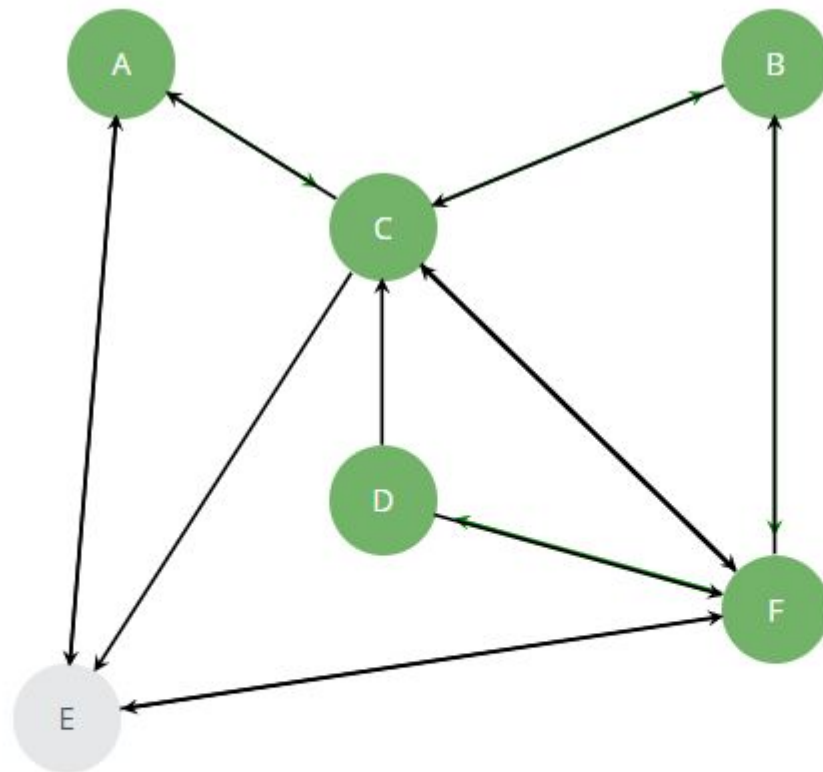
F
B
C
A

Steps:

Print (F,D) and call depth first search on D

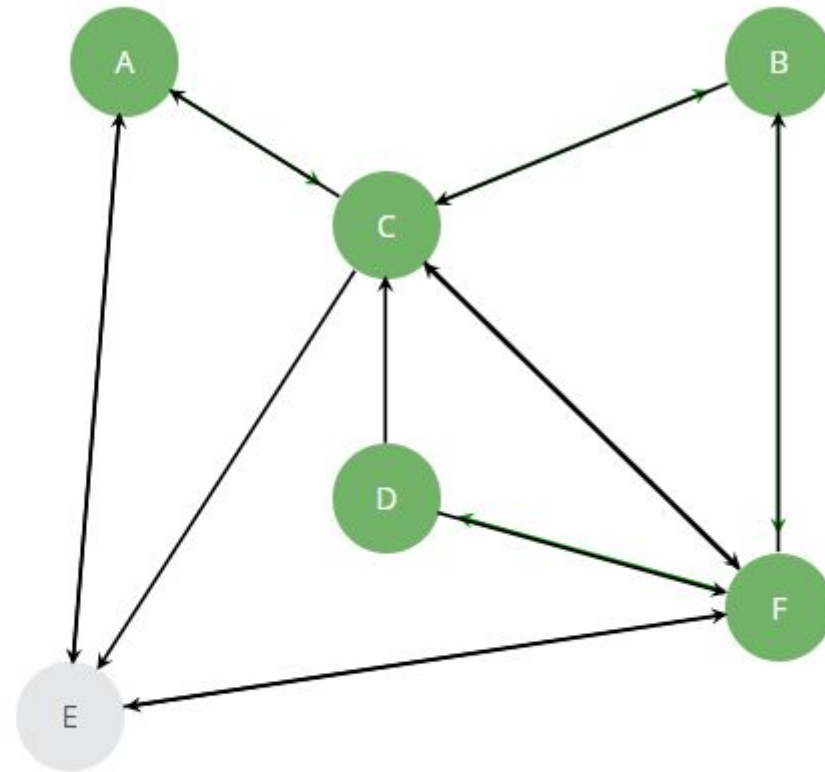


Steps:
Mark node D



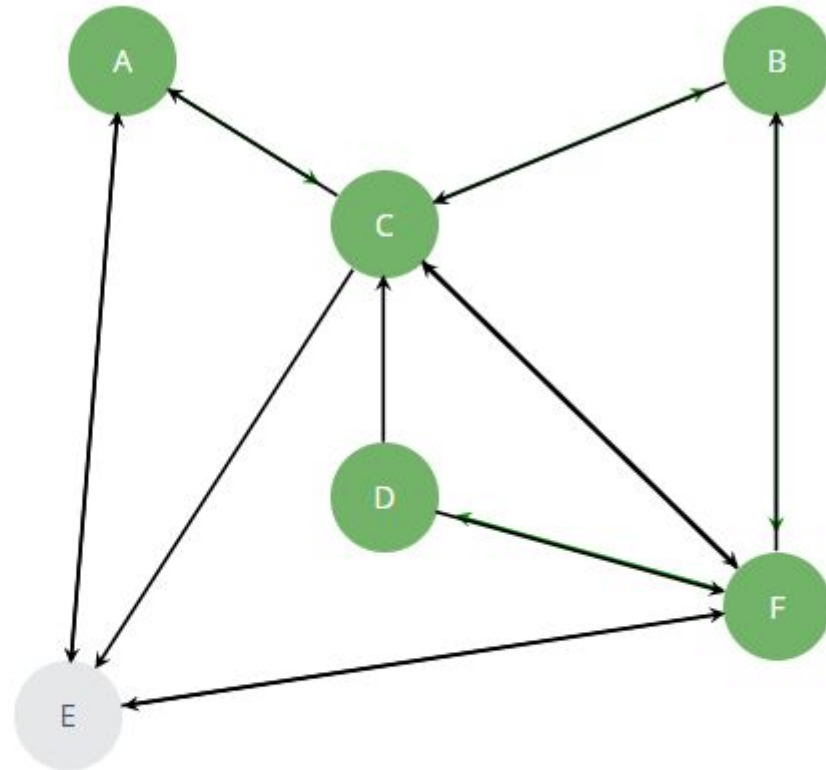
D
F
B
C
A

Steps:
Add D to the stack



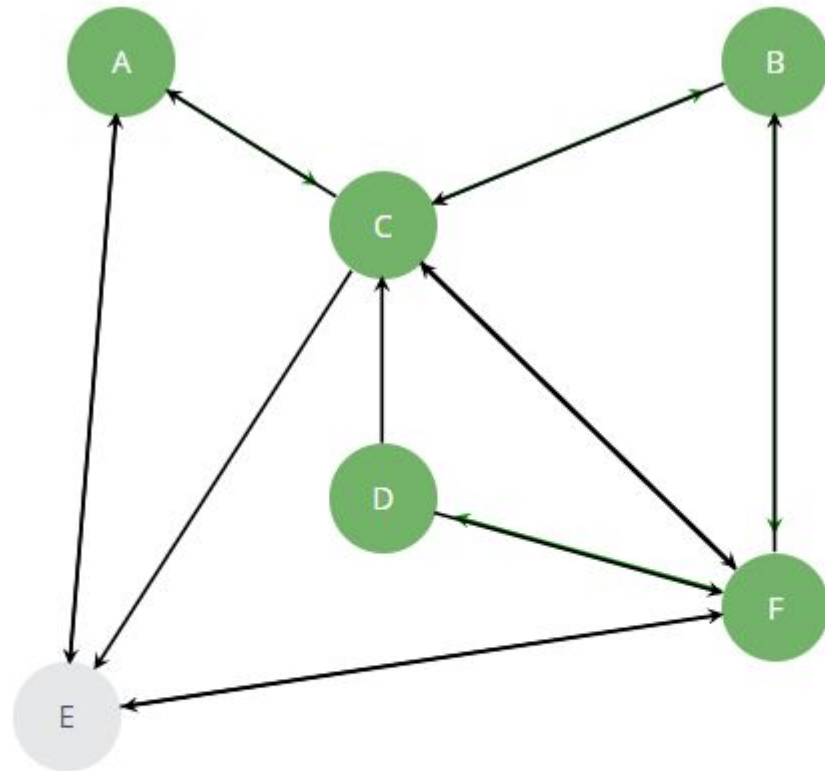
Steps:

Node C already marked

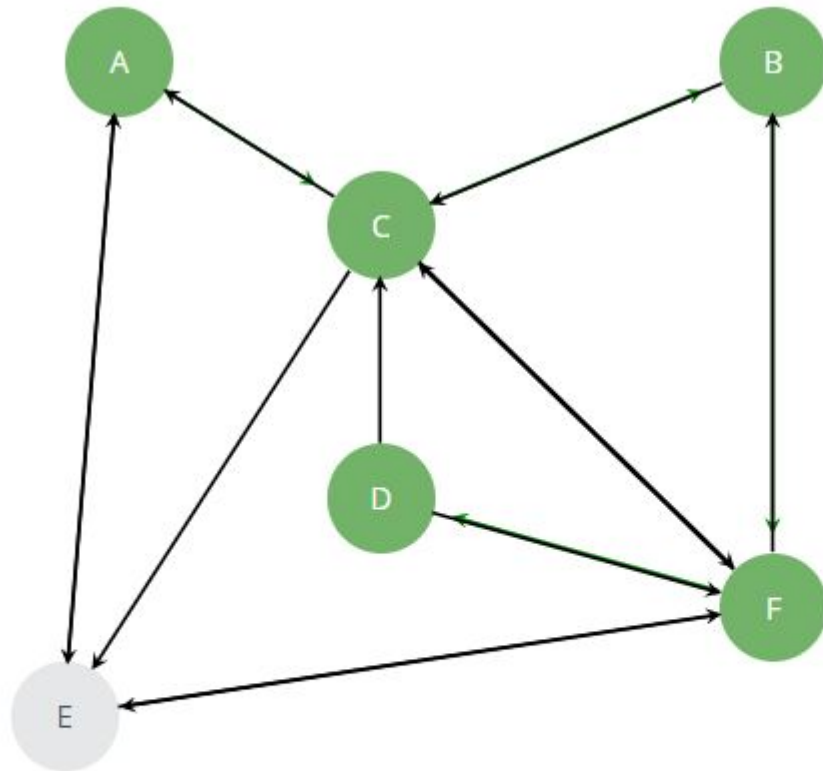


D
F
B
C
A

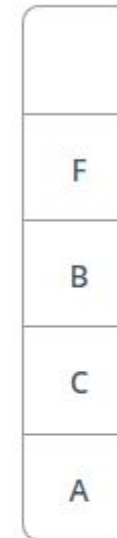
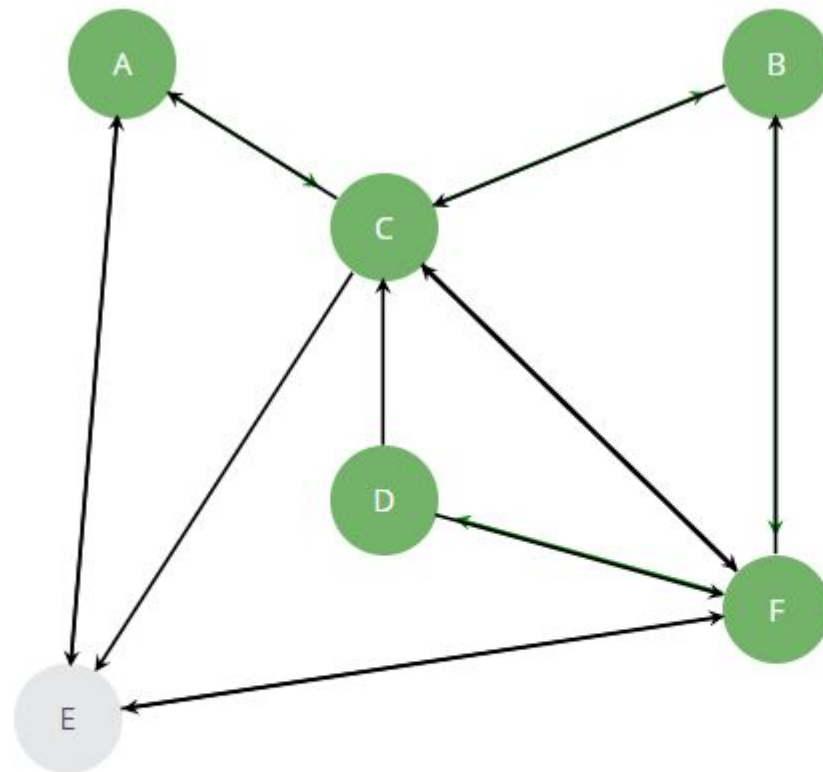
Steps:
Node F already marked



Steps:
Pop D off of stack

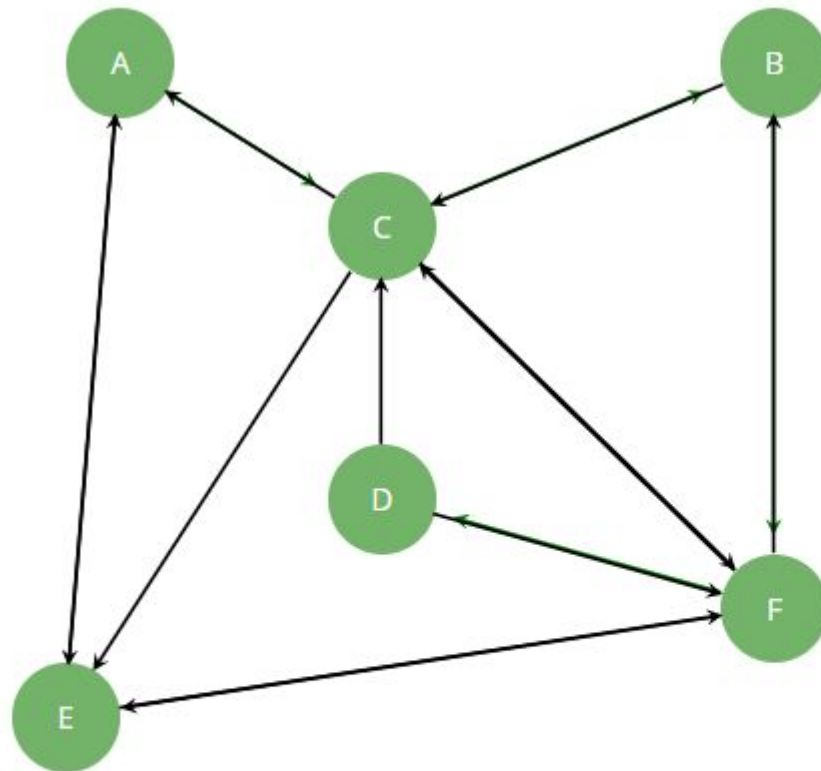


Steps:
Process (F,E)

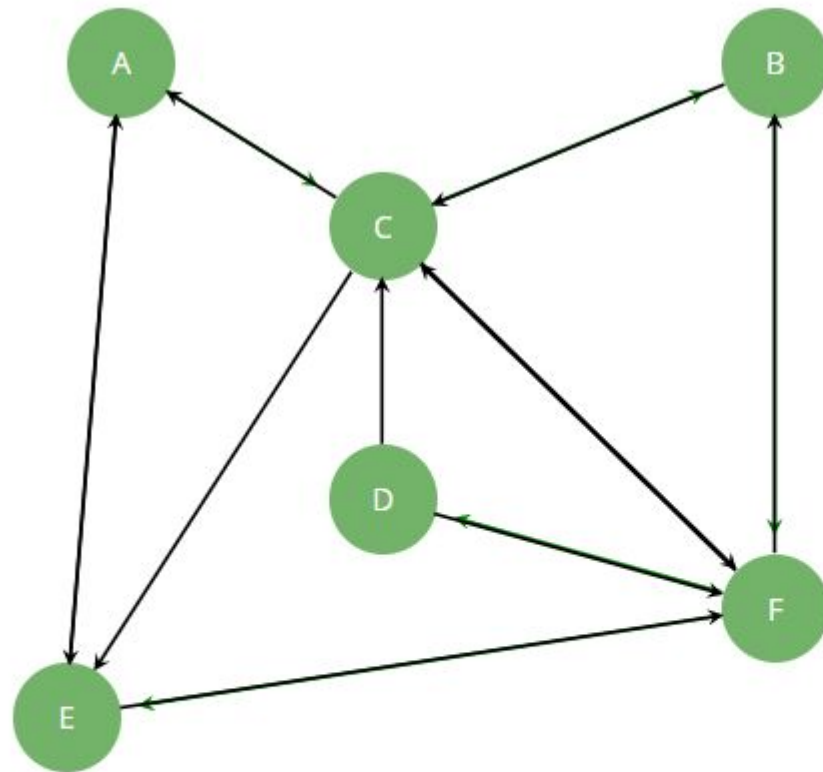


Steps:

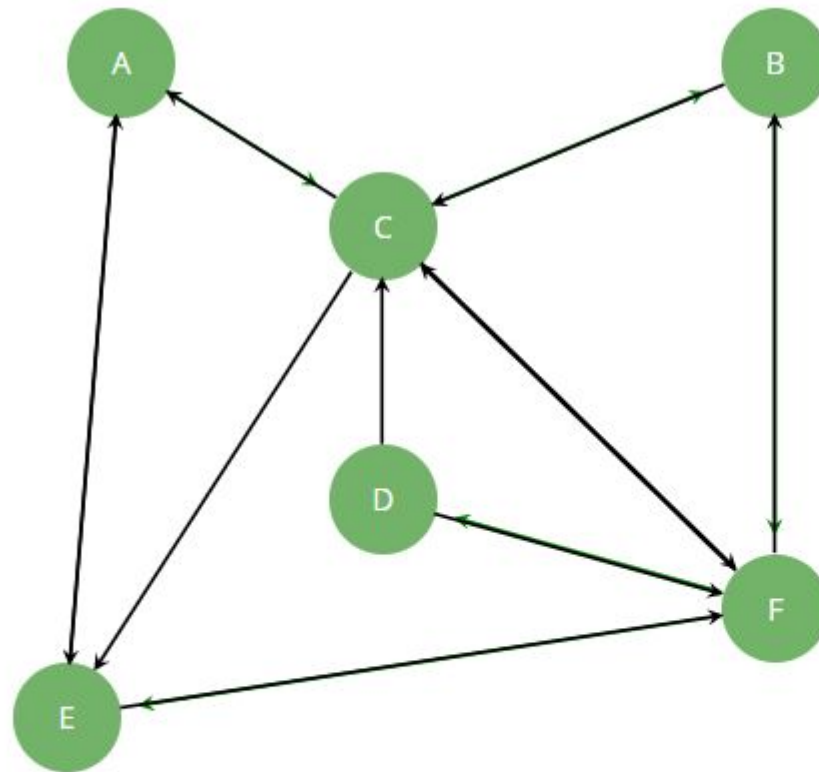
Print (F,E) and call depth first search on E



Steps:
Mark node E



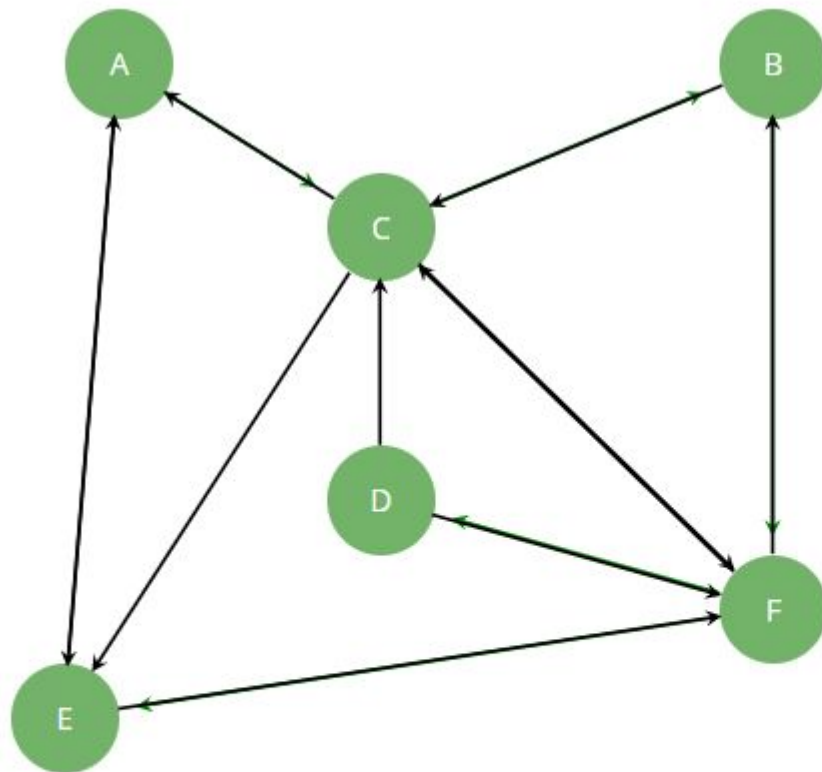
Steps:
Add E to the stack



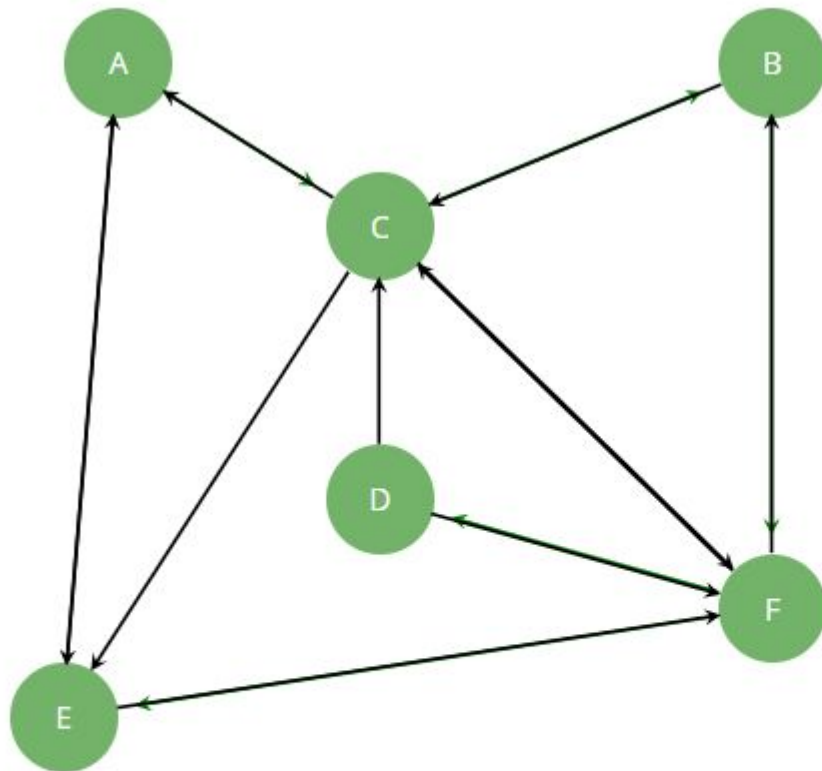
E
F
B
C
A

Steps:

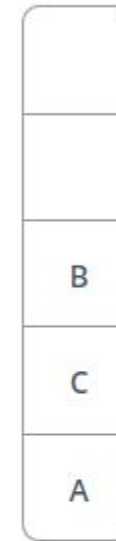
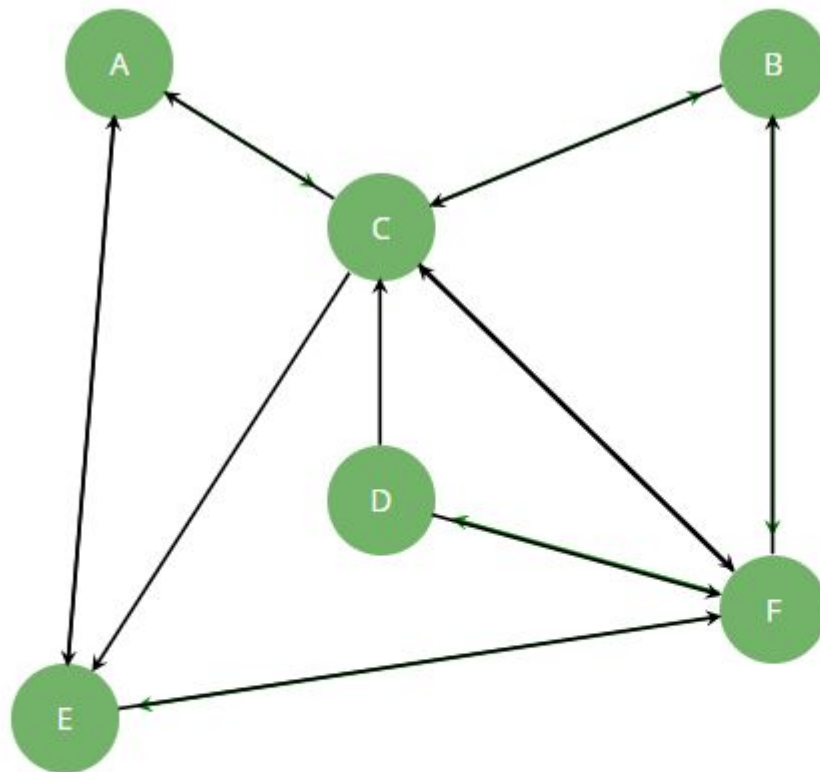
Node A already marked



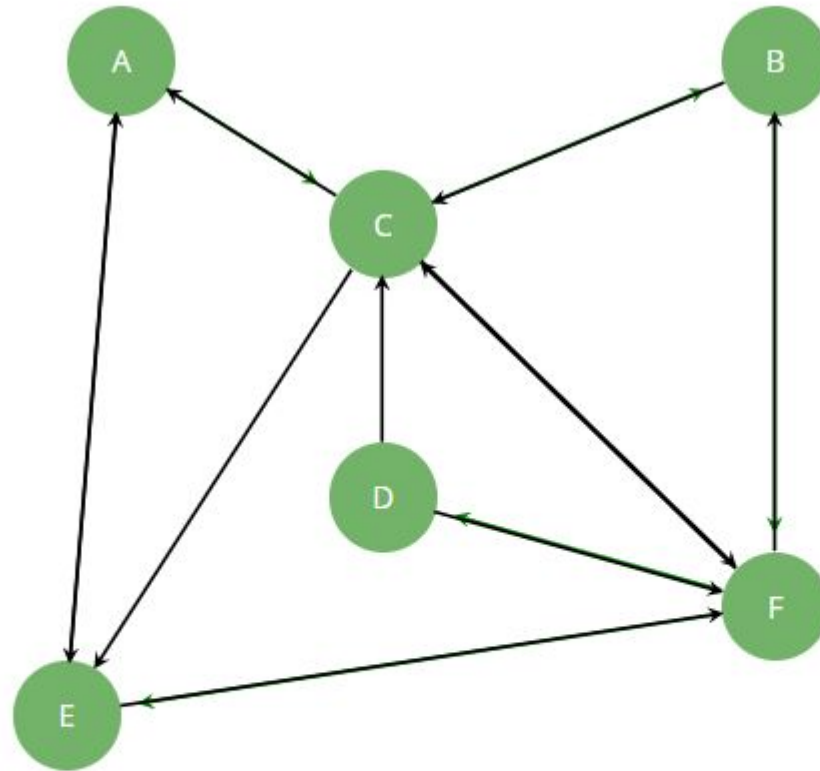
Steps:
Node F already marked



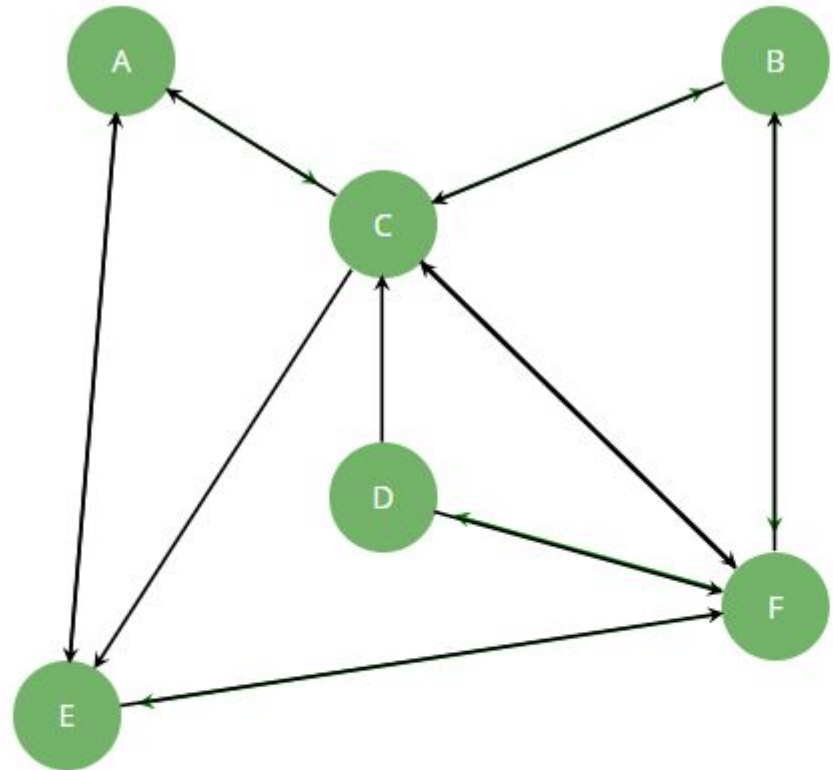
Steps:
Pop E off of stack



Steps:
Pop F off of stack

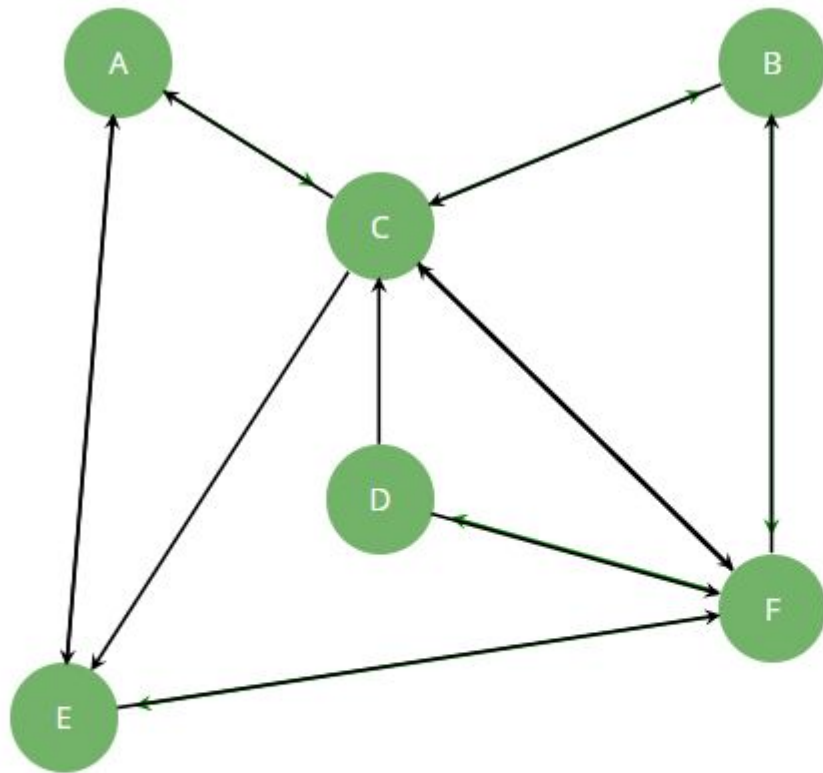


Steps:
Pop B off of stack



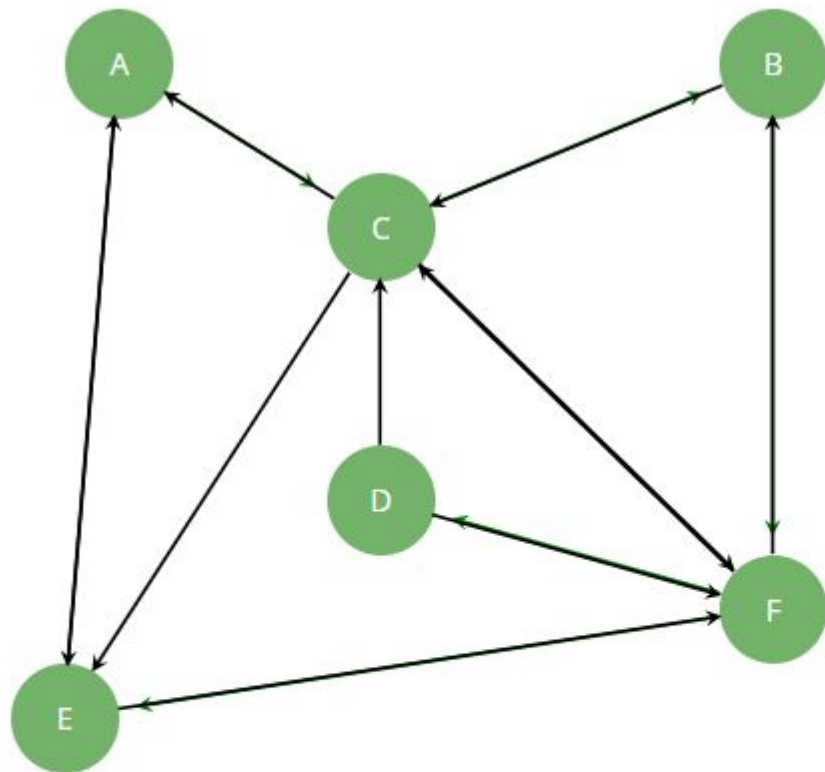
Steps:

Node E already marked

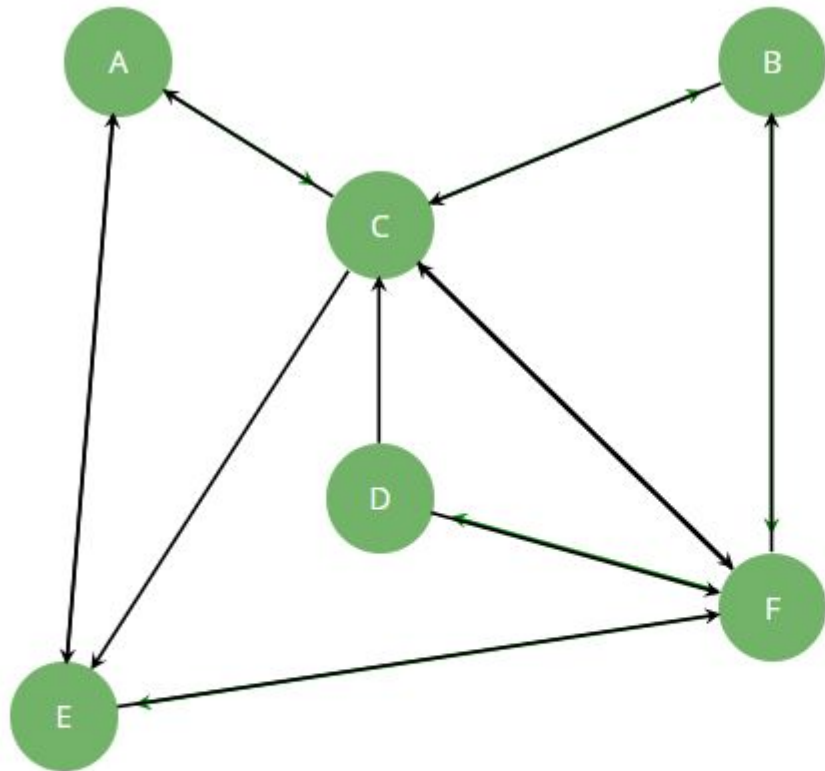


Steps:

Node F already marked

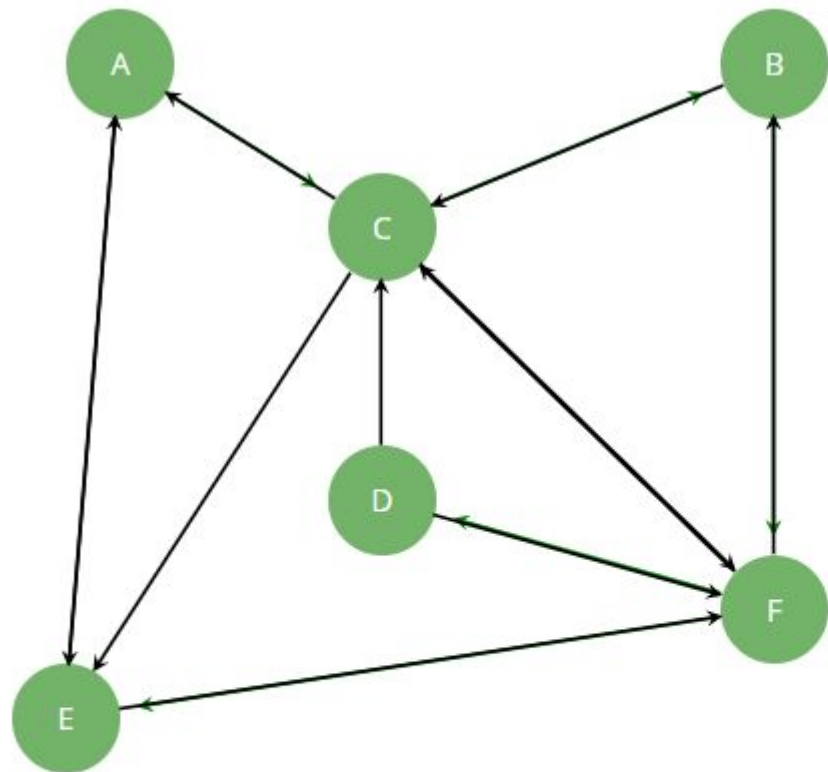


Steps:
Pop C off of stack



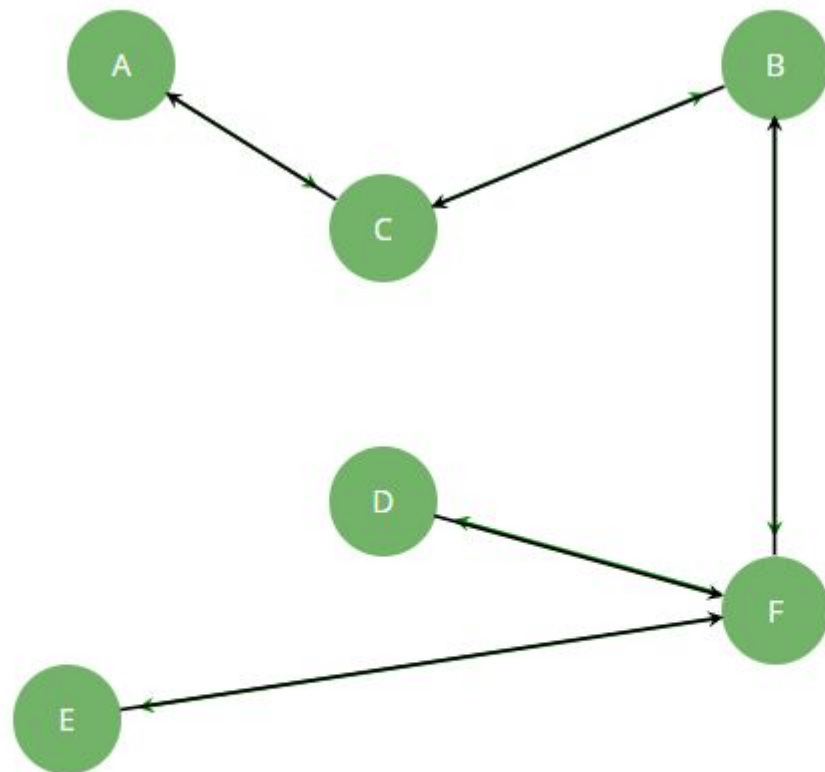
Steps:

Node E already marked



Steps:

Pop A off of stack



Steps:
Completed depth first search graph